روش‌های پیشرفته نمونه برداري در جنگل

استراتژي‌هاي نمونه برداري

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Multistage sampling

In multistage sampling, the sampling units are selected in a hierarchical order.

The population is partitioned into *primary units*, which in turn are subdivided into *secondary units*, the latter into *tertiary units*, *etc*.

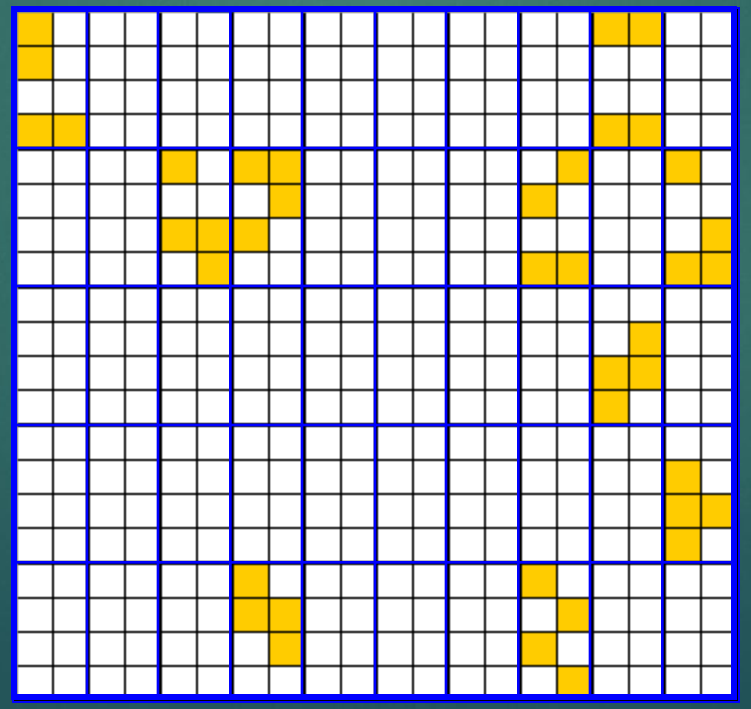
An important property of multistage sampling is the random selection of sampling units in each stage.

For example
Basin > Compartment > Parcel > Plot

Two stage sampling

In two-stage sampling, **n primary units** are drawn from a population of **N units**.
In the case of equal cluster sizes, each of the N primary units is subdivided into **M secondary units** with **m units** drawn at random within each primary unit.
The sampling fractions are n/N for the primary and m/M for the secondary units.

The total size of the sample is mn and the overall sampling fraction is mn/MN.



Estimators

Unbiased estimates of the population mean and total are

$$\bar{\bar{y}} = \frac{\sum_{m}\sum_{n} y_{ij}}{mn} \qquad Y = MN\bar{\bar{y}}$$

Estimators

An analysis of variance, which partitions the total sum of squared deviations from the overall mean into components associated with variability amongst the primary units and variability between secondary units within primary units, is required to estimate the variances involved.

Source	Sum of squares	df	Mean square
Total	$\sum^{m}\sum^{n}\left(y_{ij}-\bar{\bar{y}}\right)^2$	mn − 1	
Primary units	$m\sum^{n}\left(\bar{y}_j-\bar{\bar{y}}\right)^2$	n − 1	$\frac{SS_{\text{pu.}}}{(n_1-1)}$
Secondary units	$\sum^{n}\left(\sum^{m}\left(y_{ij}-\bar{y}_j\right)^2\right)$	N(m − 1)	$\frac{SS_{su.}}{n(m-1)}$

Estimators

In sampling from an infinite population, the expected values of the mean squares between primary units and between secondary units within primary units are

between primary units $\sigma_w^2 + m\sigma_b^2$

between secondary units σ_w^2

where

σ_b^2 = variance amongst primary units

σ_W^2 = variance amongst secondary units

The variance of the overall mean

$$\sigma_{\bar{y}}^2 = \frac{m\sigma_b^2 + \sigma_w^2}{mn} = \frac{\sigma_b^2}{n} + \frac{\sigma_w^2}{mn}$$

Sample allocations

The allocation of sampling units requires **prior knowledge of the variance** amongst primary and secondary units, respectively.

Two cost components are involved:
- Sampling cost per primary unit *(c1)*
- Sampling cost per secondary unit *(c2)*

The cost element *c*1 represents the cost of traveling from one primary unit to the second, whereas *c*2 gives the cost of measuring the secondary unit.
The optimum allocation rule prescribes the number of secondary sampling units to be measured:

$$m = \sqrt{\frac{\sigma_w^2}{\sigma_b^2} \cdot \frac{c_1}{c_2}}$$

Sample allocations

The number of primary units is given by

$$n = \frac{\sigma_b^2 + \frac{\sigma_w^2}{m}}{\left[s_{\bar{\bar{y}}}^2 + \frac{1}{N} \cdot \left(\sigma_b^2 + \frac{\sigma_w^2}{M}\right)\right]} \text{ with } s_{\bar{\bar{y}}}^2 \text{ being specified.}$$

σ_b^2 = variance amongst primary units

σ_W^2 = variance amongst secondary units

$t \cdot s_{\bar{\bar{y}}}$

Three stage sampling

The basic principle of two-stage sampling may be extended to multistage sampling.

When three stages are involved,

n primary units are drawn at random from a population of *N* units,
m secondary units from a population of *M* subunits per primary unit
k tertiary units from a population of *K* units per secondary unit.

The total sample size is *nmk* and represents a sampling fraction of *nmk/NMK*.

Three stage sampling

In a study to estimate the mean nitrogen content of the foliage of trees within a given stand

a primary sample of n trees is drawn at random from a population of N units,
a subsample of m branches is selected from a population of M branches in each tree
a subsample of k leaves within each of these branches, from a population of K leaves in each branch.

We assume that N, M, and K are known, and each tree has the same number of branches and each branch the same number of leaves per branch. The resultant sample fractions are
Trees $f_1 = n/N$
Branches $f_2 = m/M$
Leaves $f_3 = k/K$

Three stage sampling

	$E(MS)$
Between trees	$\sigma_l^2 + k \cdot \sigma_{br}^2 + m \cdot k \cdot \sigma_{tr}^2$
Between branches (within trees)	$\sigma_l^2 + k \cdot \sigma_{br}^2$
Between leaves (within branches)	σ_l^2

Where σ_{tr}^2 = variance among trees, σ_{br}^2 = variance among branches (within trees) and σ_l^2 = variance among leaves (within branches). The variance of the population mean, corrected for the finite population is

$$s_{\bar{\bar{y}}}^2 = \left(\frac{1}{nmk}\right)^2 \left[(1 - f_1) \cdot s_{tr}^2 + f_1 \cdot (1 - f_2) \cdot s_{br}^2 + (1 - f_3) \cdot f_1 \cdot f_2 \cdot s_l^2\right].$$

The decision of how to distribute the sample amongst stands, trees and branches, however, is also affected by cost considerations.

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